13.5 Summary

- **Canopy clustering**: an algorithm to compute a coarse-grained pseudo-clustering of a dataset, with the advantage of being very inexpensive, compared to more accurate clustering algorithms like k-means or DBSCAN. It can be used as part of k-means, replacing the random centroids initialization step, or to coarsely shard a dataset, in distributed computing.
- **Parallel computing**: for some algorithms, parallelizing the execution can guarantee a dramatic reduction of running time and memory needed. We distinguish between parallel computing, running some software in several threads on the same machine, and distributed computing, where several machines, and possibly resources on the cloud, are used to run a software in a joint effort.
- **MapReduce**: a computational model that leverages the cloud to scale out processing of large datasets. It mainly requires 3 things: sharding, a mapper phase that transforms data in each shard independently, and a reduce phase that gathers data from mappers according to some criteria, and combine them.
- **MapReduce-based clustering**: k-means, canopy clustering and DBSCAN can all be re-written as distributed algorithms by using the MapReduce model. This allows processing huge datasets that wouldn’t fit into a single machine’s memory, or that would require a huge time to be processed on any super-computer.